Capítulo 6 - Perspectivas

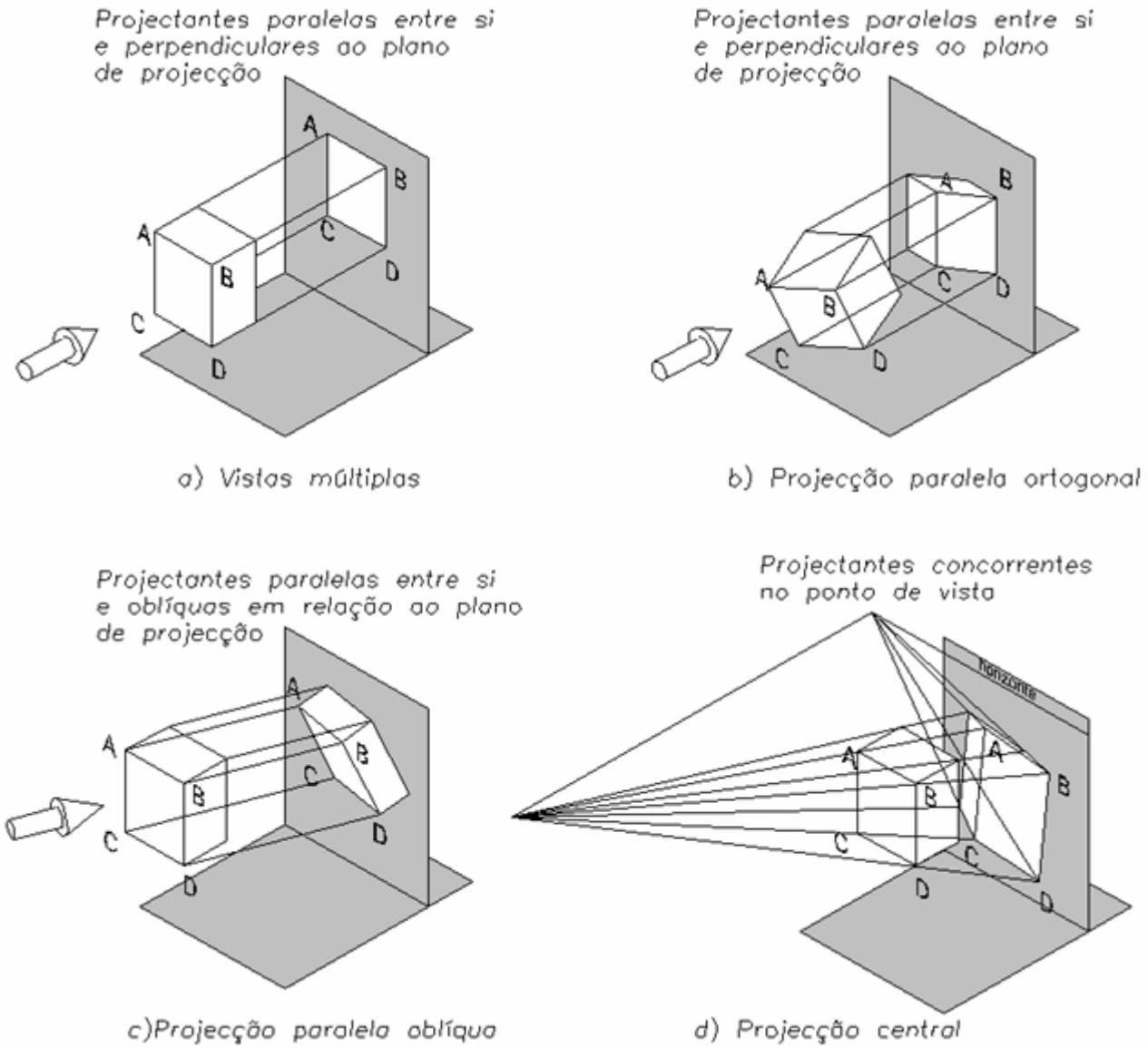
OBJECTIVOS

- Descrever as diferenças, vantagens e desvantagens entre a representação em vistas múltiplas, em projecções oblíquas, em perspectiva e em projecções centrais;
- Representar planos inclinados e círculos em perspectivas isométricas;
- Desenhar rigorosamente a perspectiva ou projecção oblíqua de qualquer objecto;
- Desenhar uma perspectiva de um objecto partindo da sua representação em vistas múltiplas;
- Esboçar à mão livre uma perspectiva de um objecto;

Silva, Ribeiro, Dias, Sousa

PERSPECTIVAS

Diferenças fundamentais entre alguns tipos de projecção



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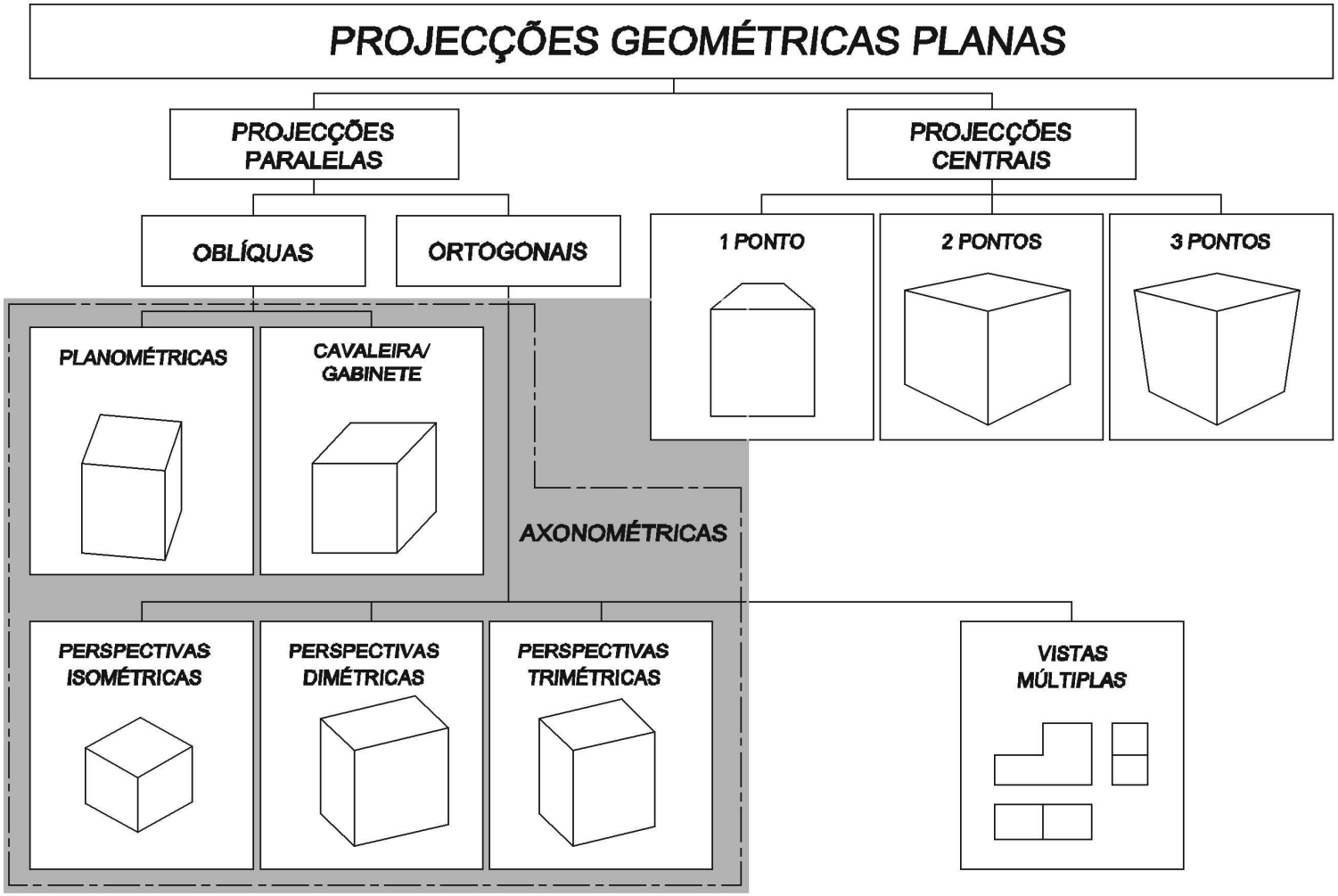
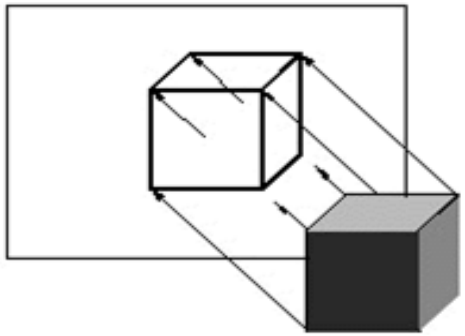
PERSPECTIVAS

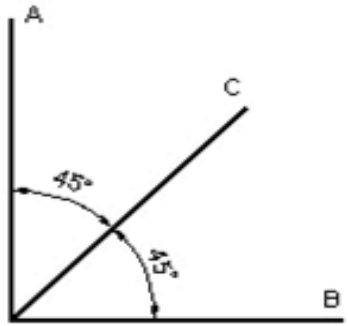
- Tipos de projecções geométricas planas

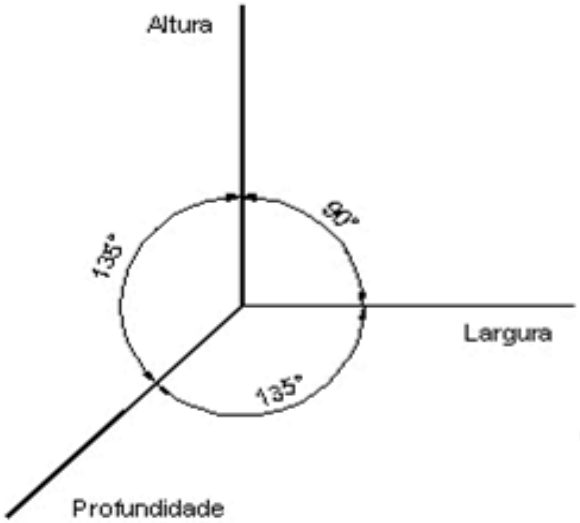


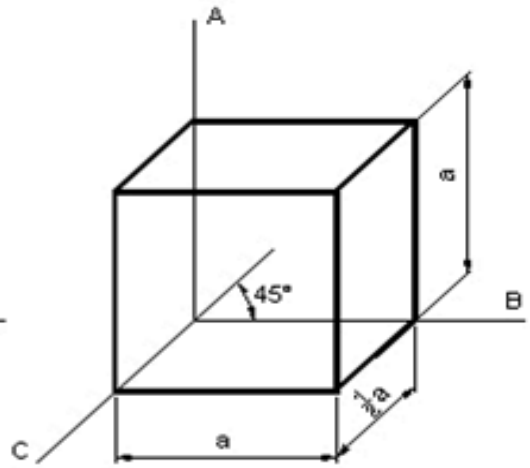
PERSPECTIVAS

- Obtenção da perspectiva de gabinete



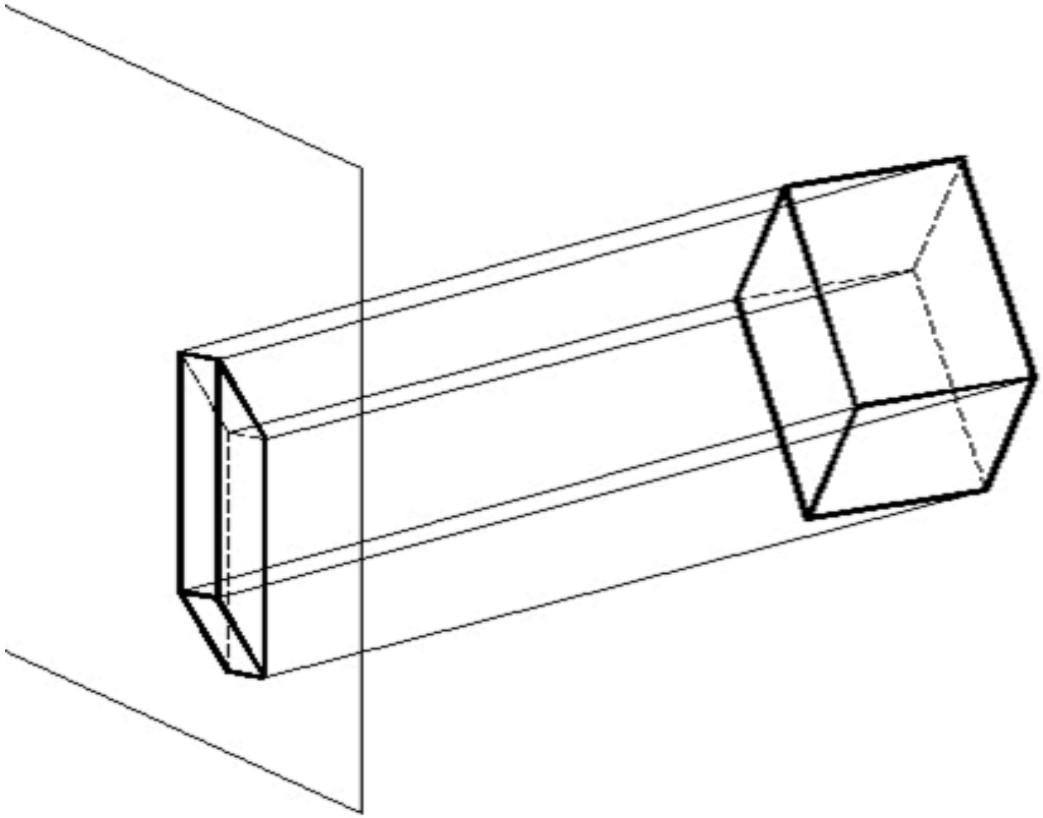






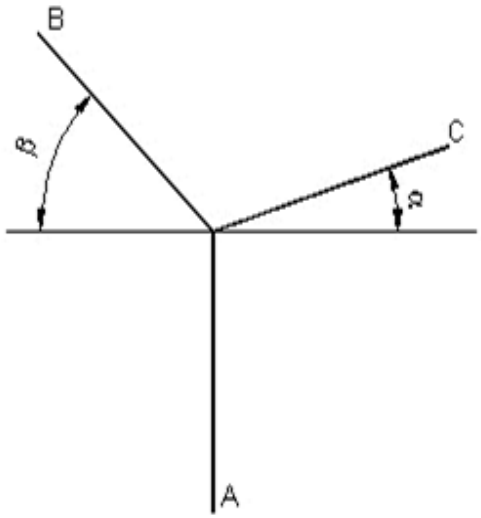
PERSPECTIVAS

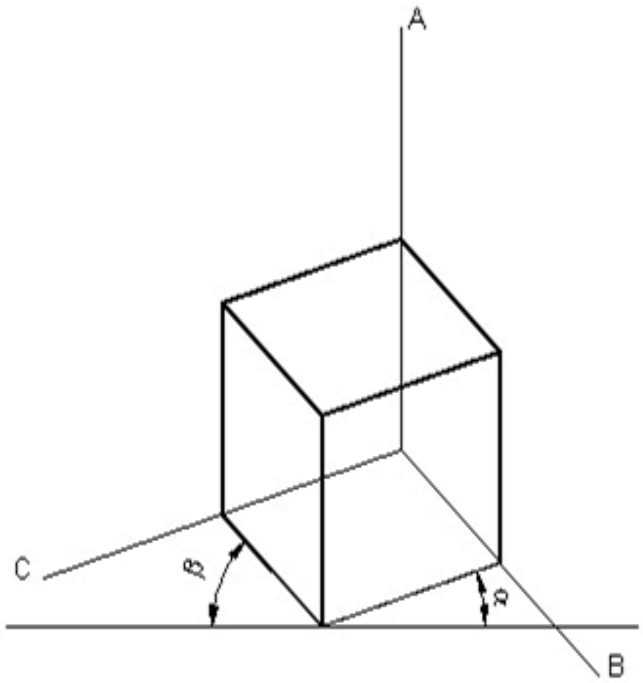
- Posição possível do objecto para obtenção de perspectiva trimétrica



PERSPECTIVAS

- Obtenção da perspectiva trimétrica





ângulo α	ângulo β	eixo A	eixo B	eixo C
5º 10’	17º 50’	1	0.9	0.5
9º 50’	24º 30’	1	0.9	0.6
14º 30’	26º 40’	1	0.9	0.7
11º 50’	16º	1	0.9	0.7

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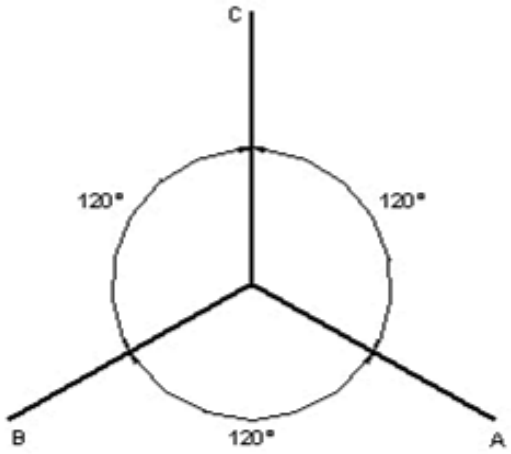
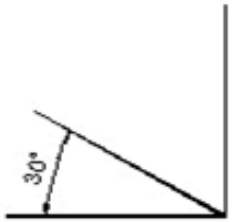
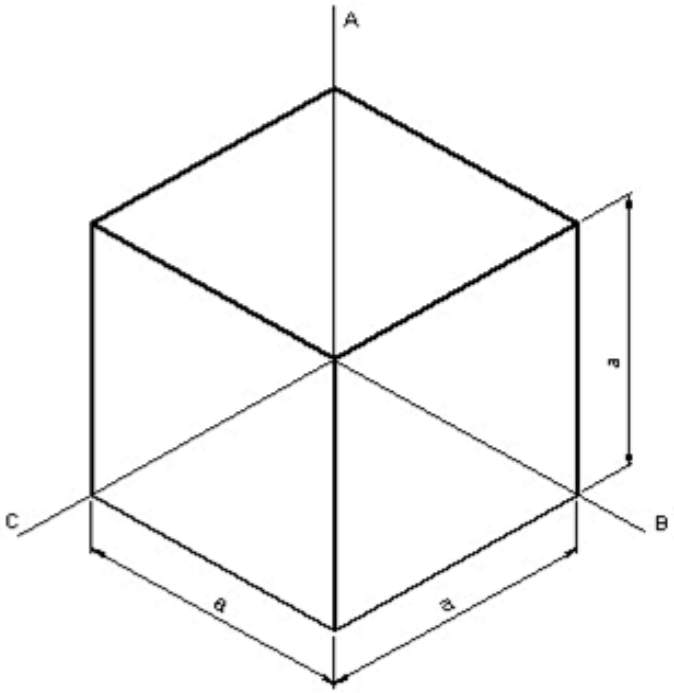
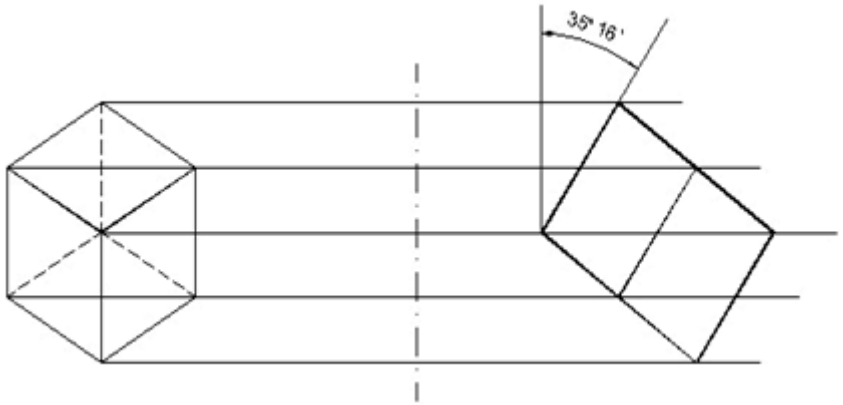
PERSPECTIVAS

- Obtenção de perspectiva isométrica









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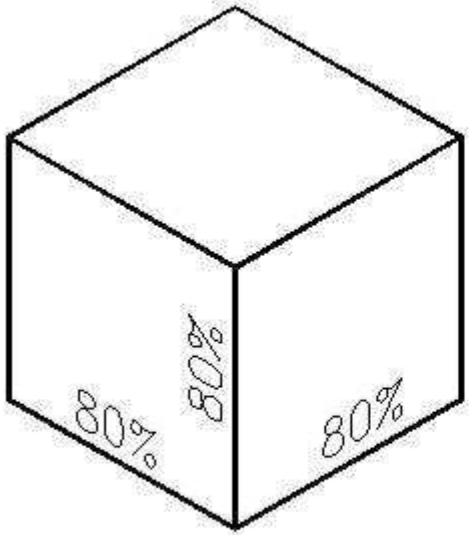
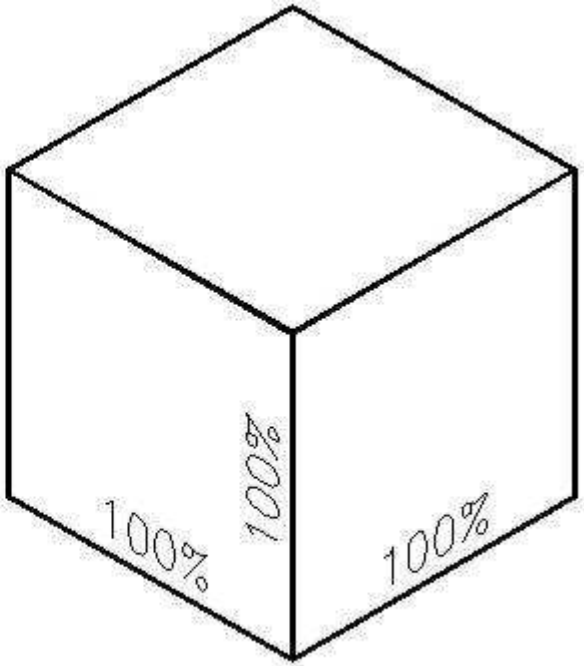
PERSPECTIVAS

Isométrica real e isométrica simplificada



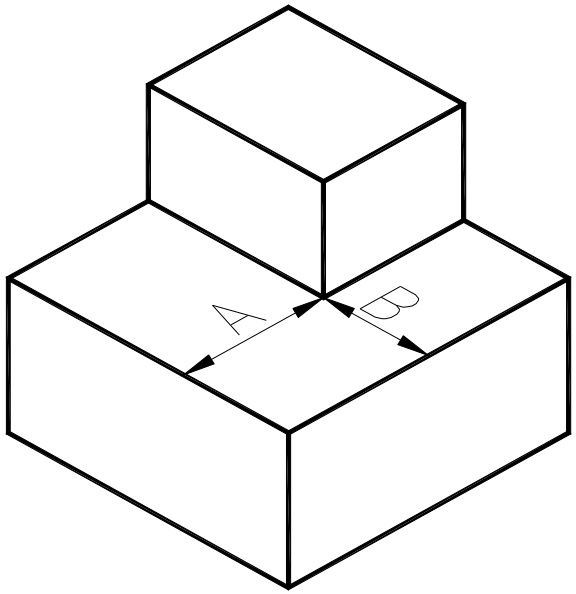
Isométrica real

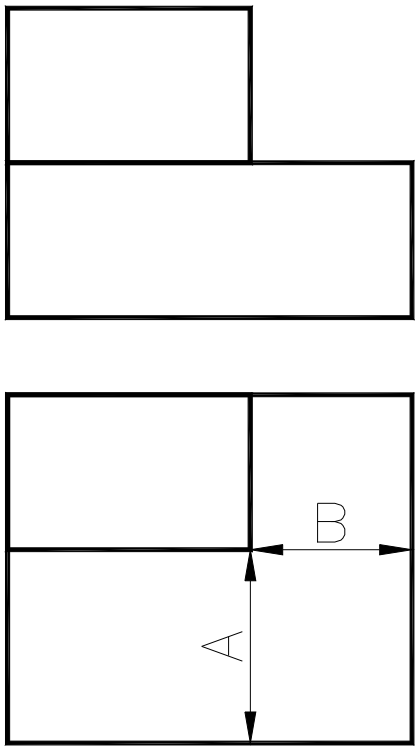


Isométrica simplificada

PERSPECTIVAS

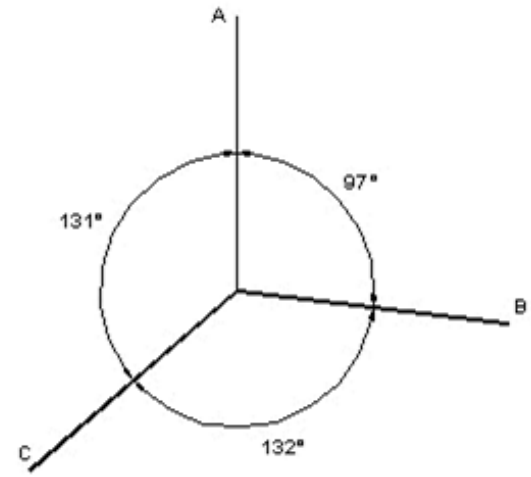
- Medições em perspectiva

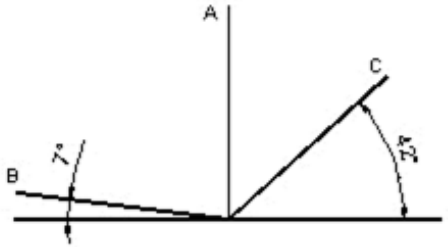


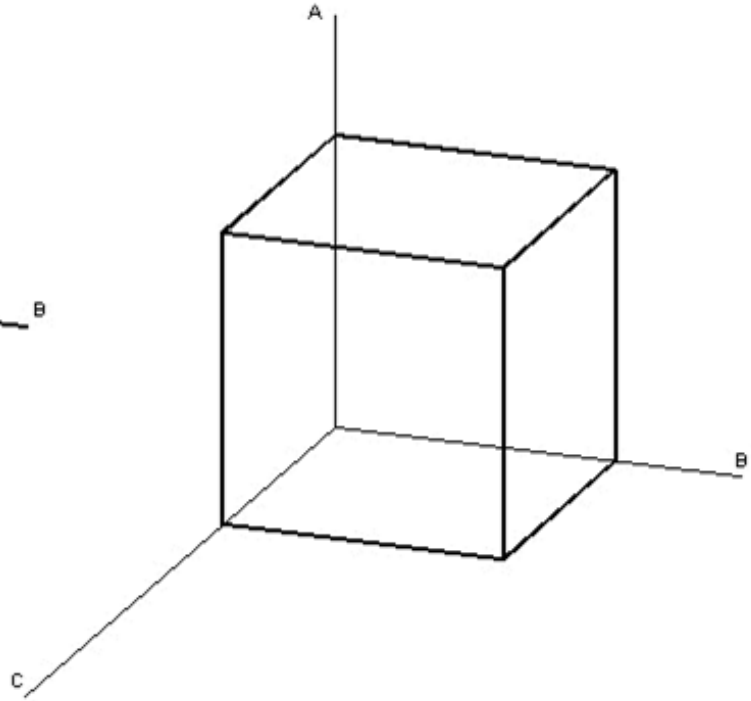


PERSPECTIVAS

- Obtenção de perspectiva dimétrica







ângulo α	ângulo β	eixo A	eixo B	eixo C
7º	42º	1	1	0.5
10º 22’	39º 49’	1	1	0.6
14º 10’	37º 55’	1	1	0.7
18º 40’	35º 40’	1	1	0.7

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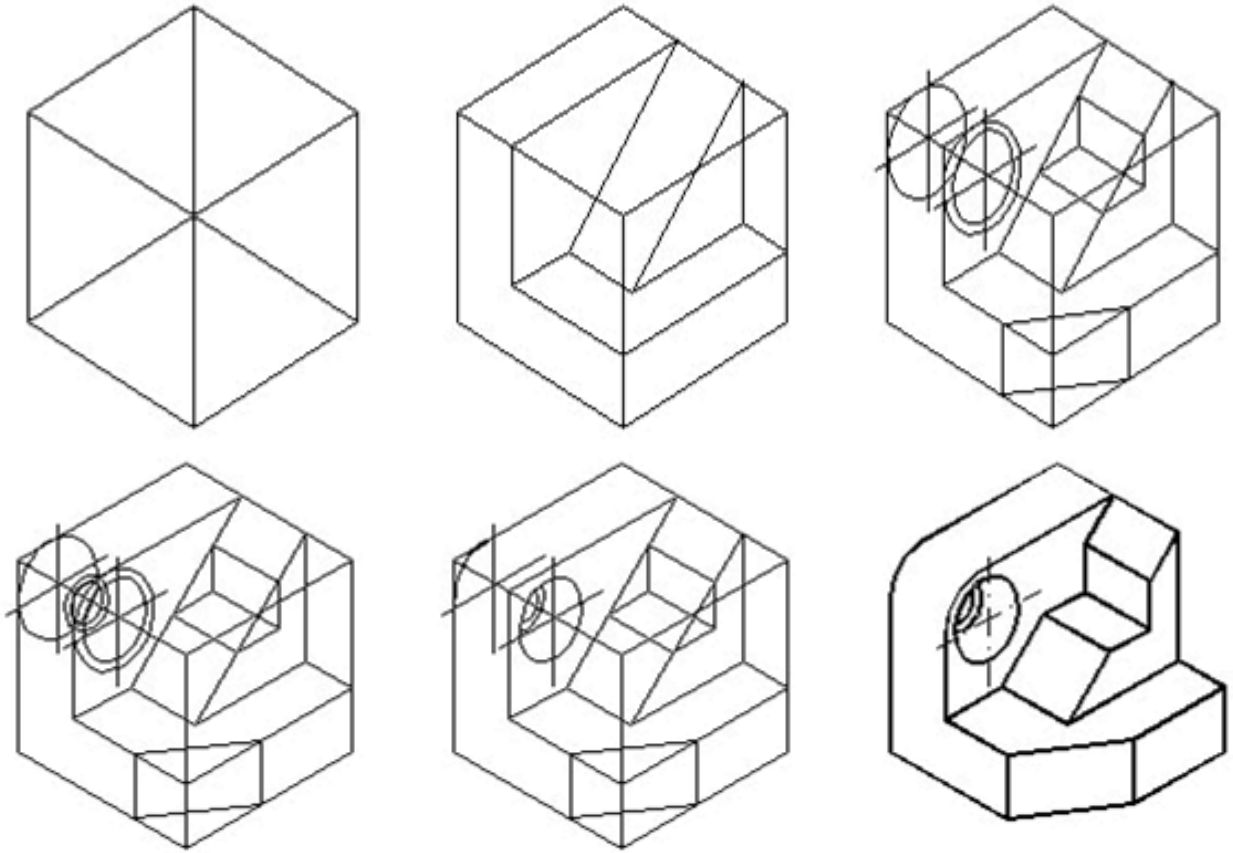
PERSPECTIVAS

- Sequência para a obtenção da representação em perspectiva de uma peça



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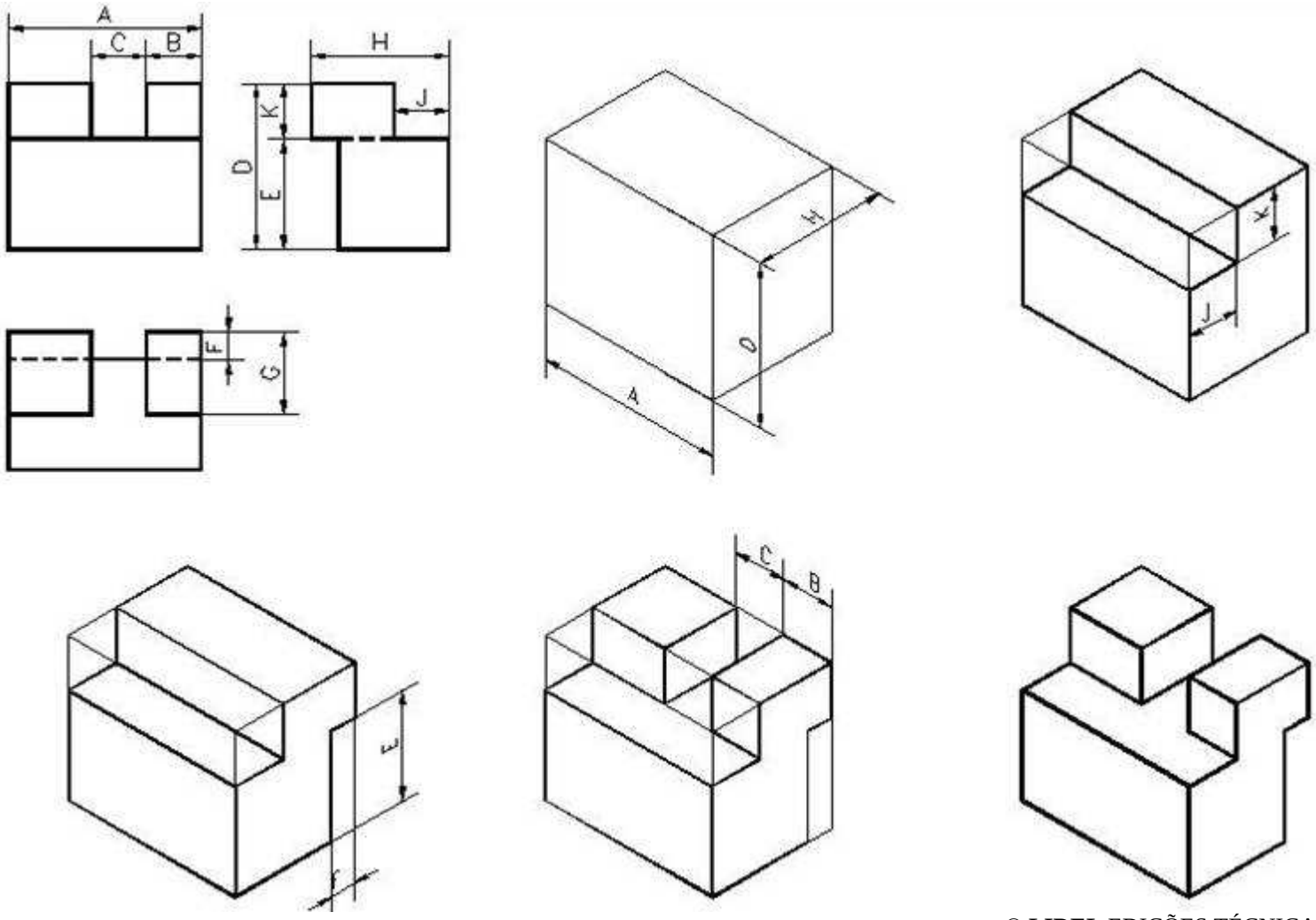
PERSPECTIVAS

Construção de objectos em perspectiva isométrica

Medições tomadas apenas segundo os eixos isométricos !



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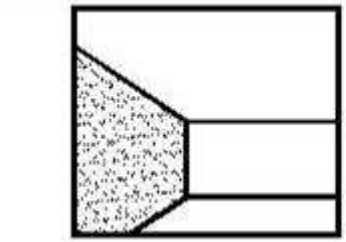
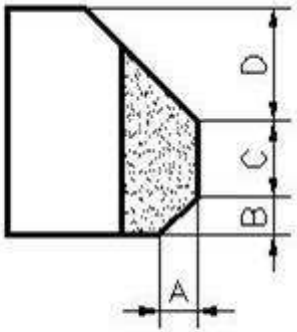
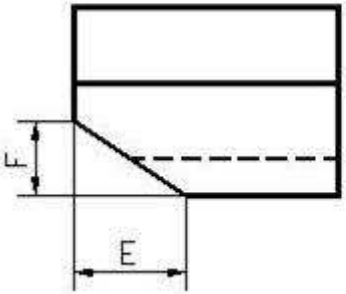
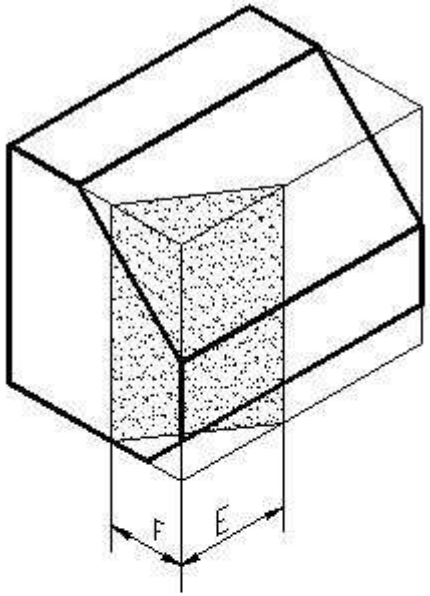
PERSPECTIVAS

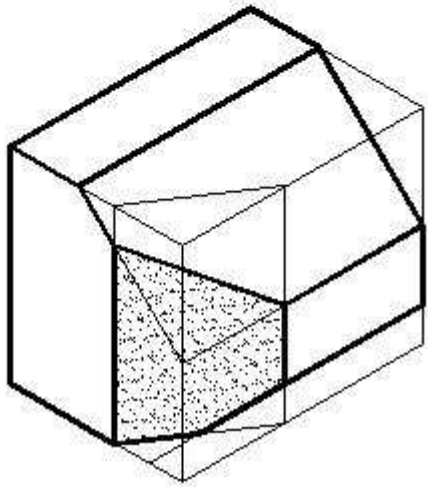
- Construção de planos inclinados





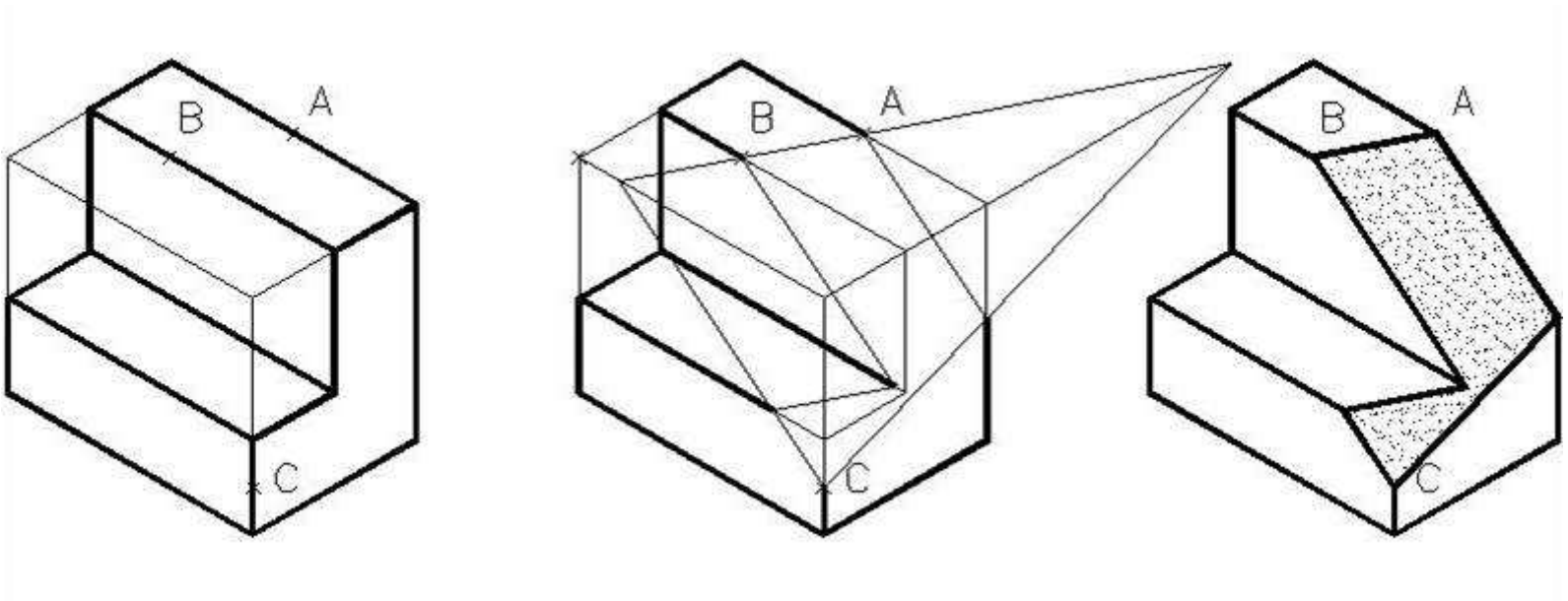






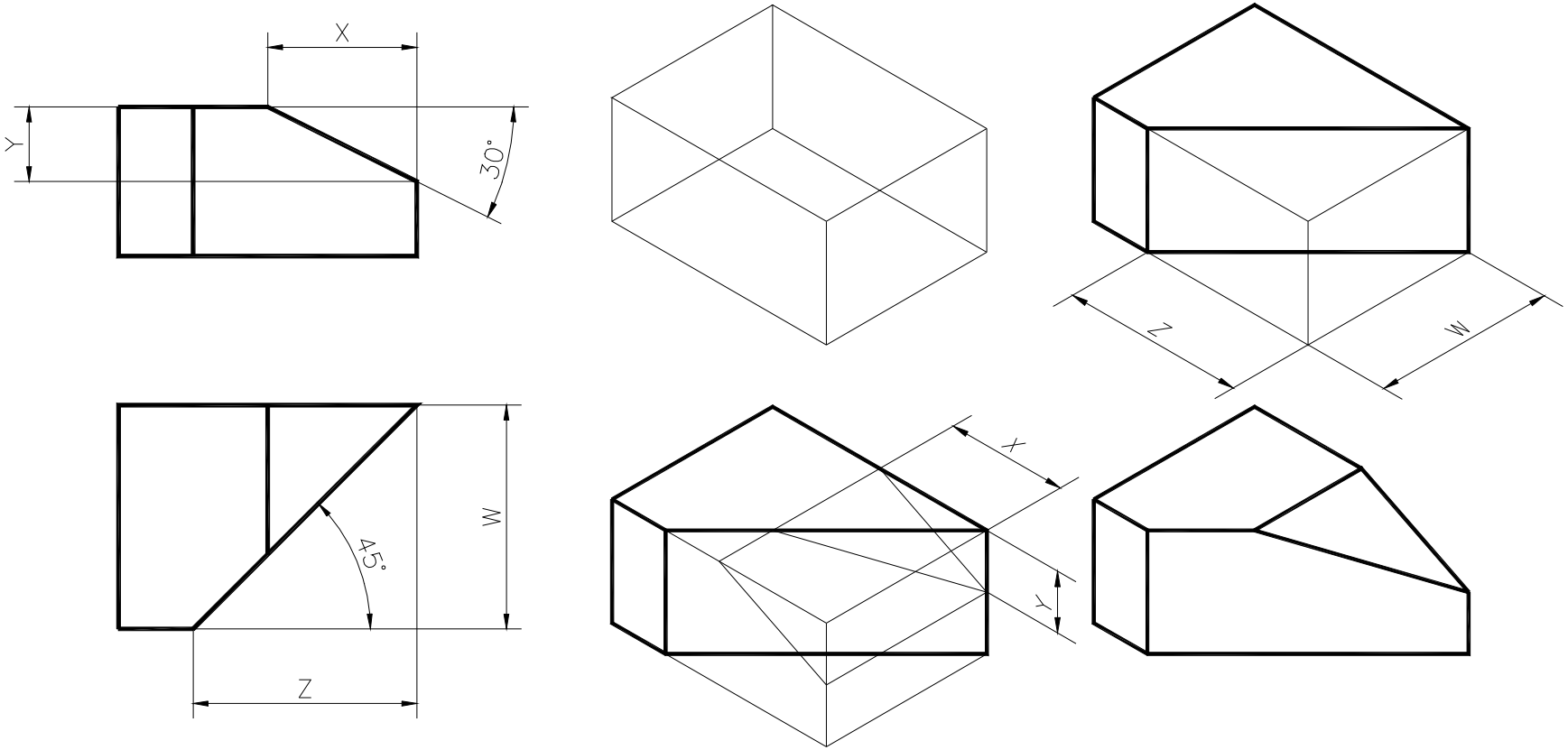
PERSPECTIVAS

- Construção de planos oblíquos



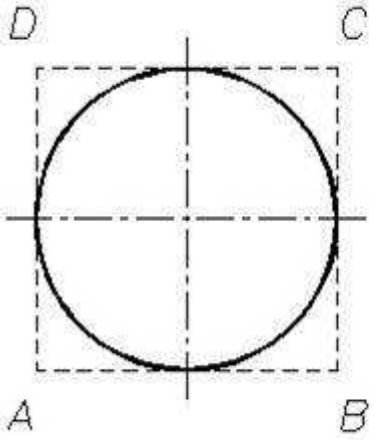
PERSPECTIVAS

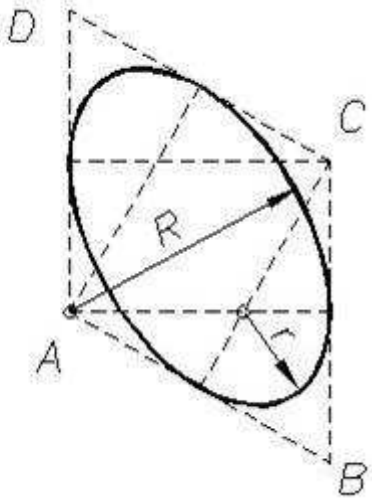
- Marcação e construção de ângulos

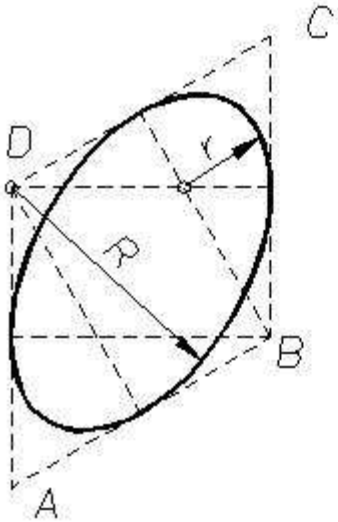


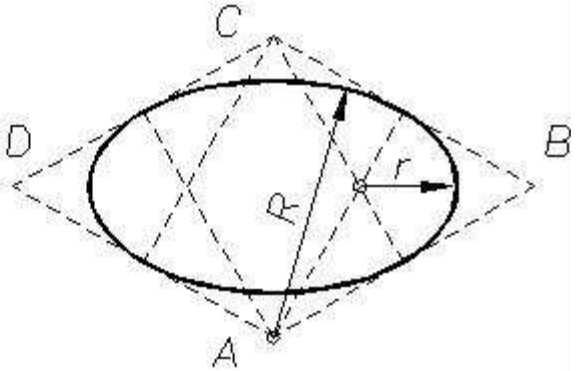
PERSPECTIVAS

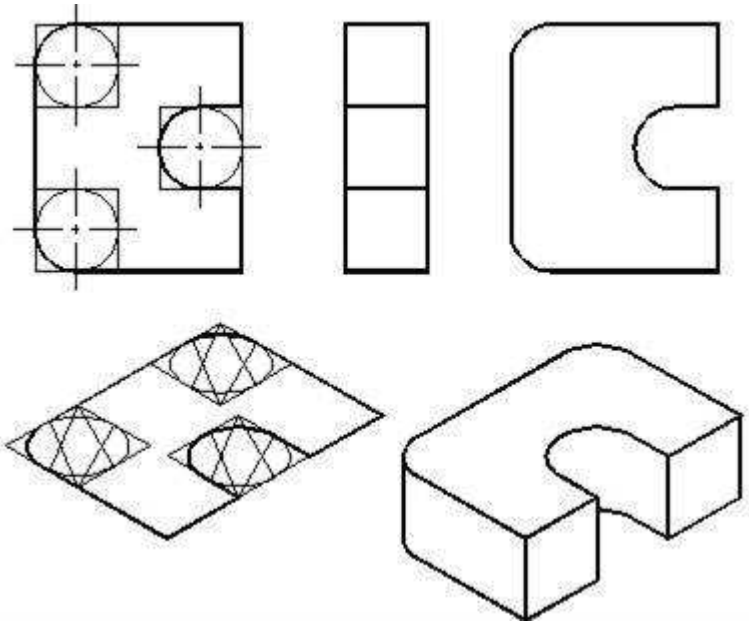
- Construção de circunferências





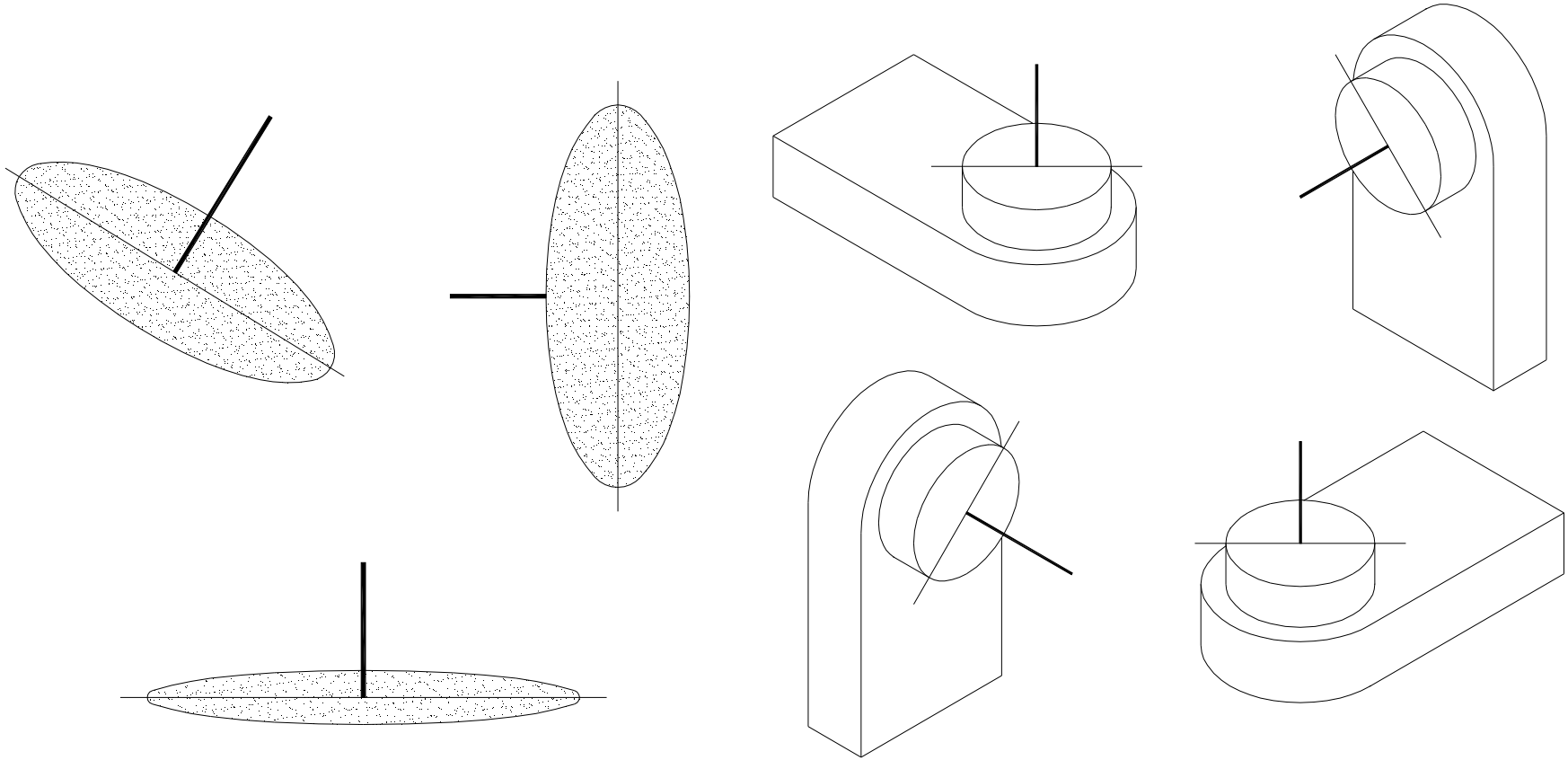






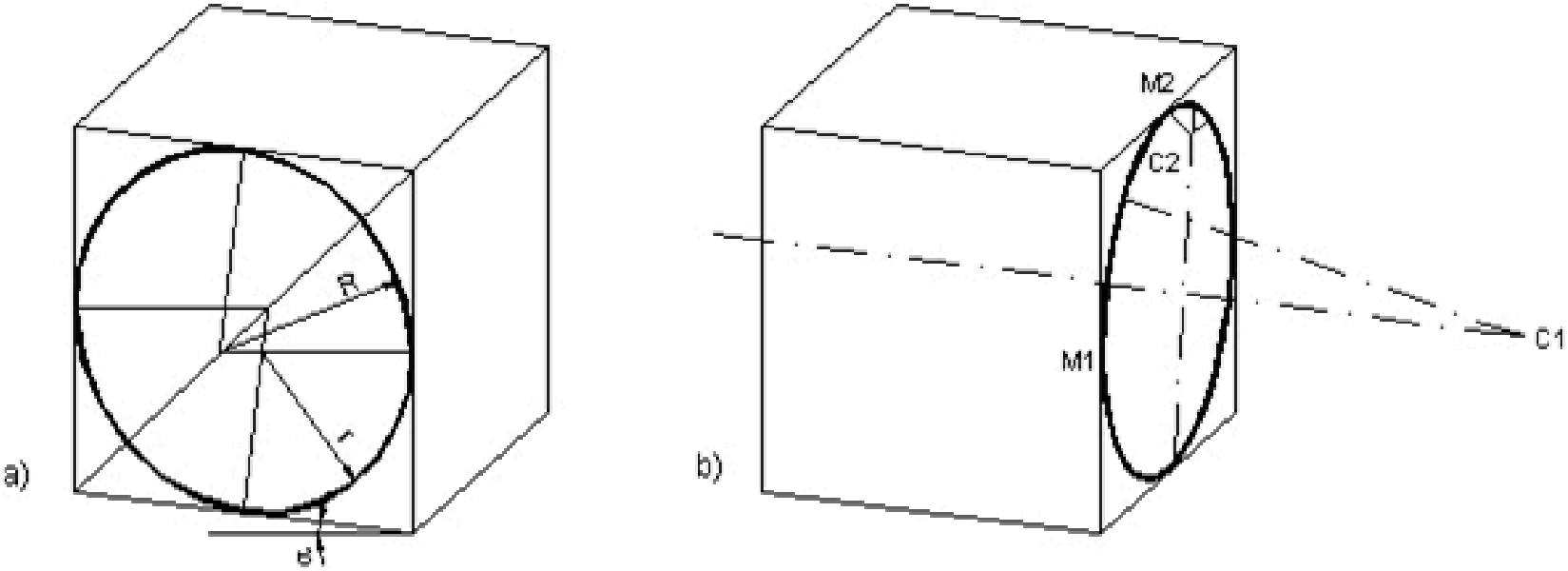
PERSPECTIVAS

Construção de elipses (e circunferências) em qualquer plano oblíquo



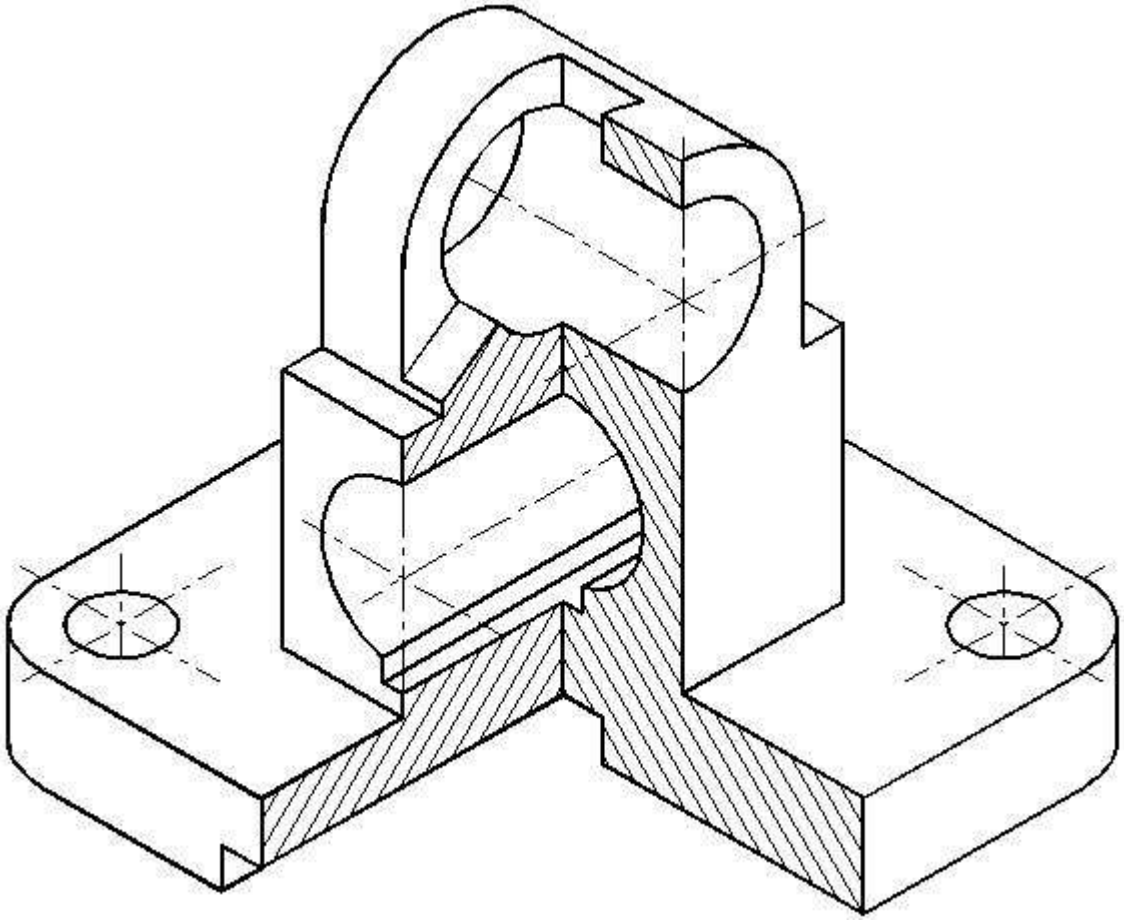
PERSPECTIVAS

- Traçado de circunferência em dimetria



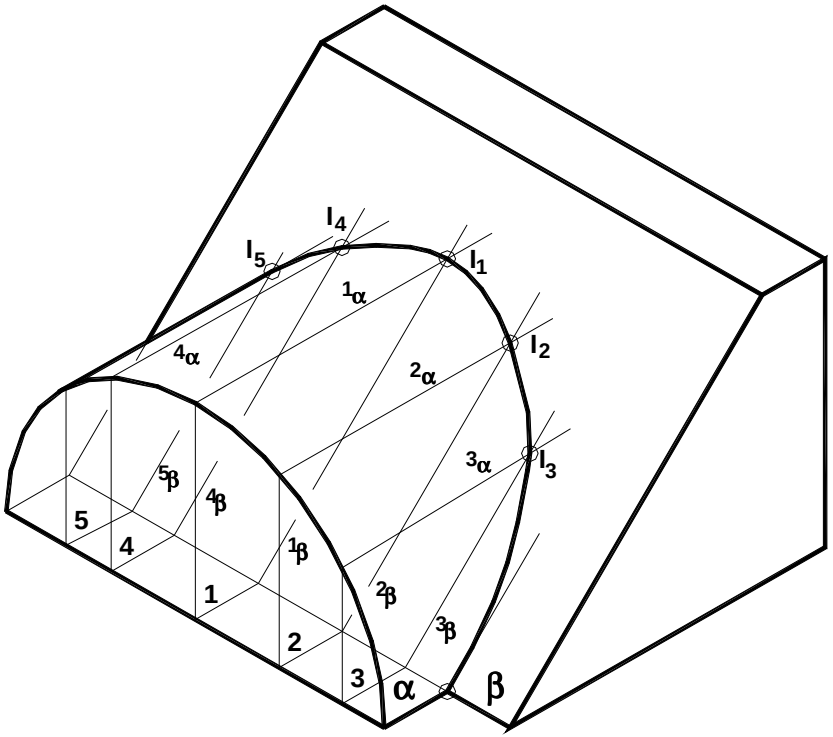
PERSPECTIVAS

Cortes em perspectiva



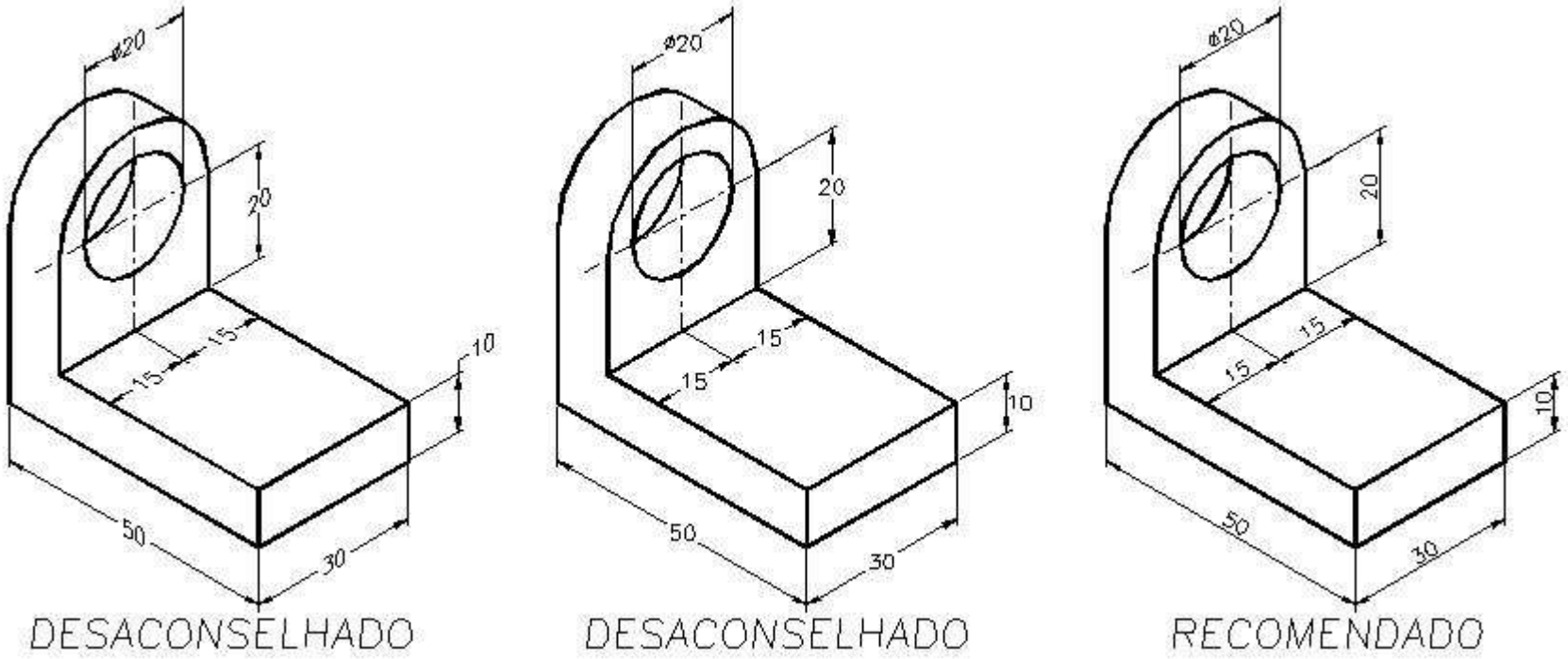
PERSPECTIVAS

- Determinação da intersecção de duas faces sendo uma delas uma superfície não plana



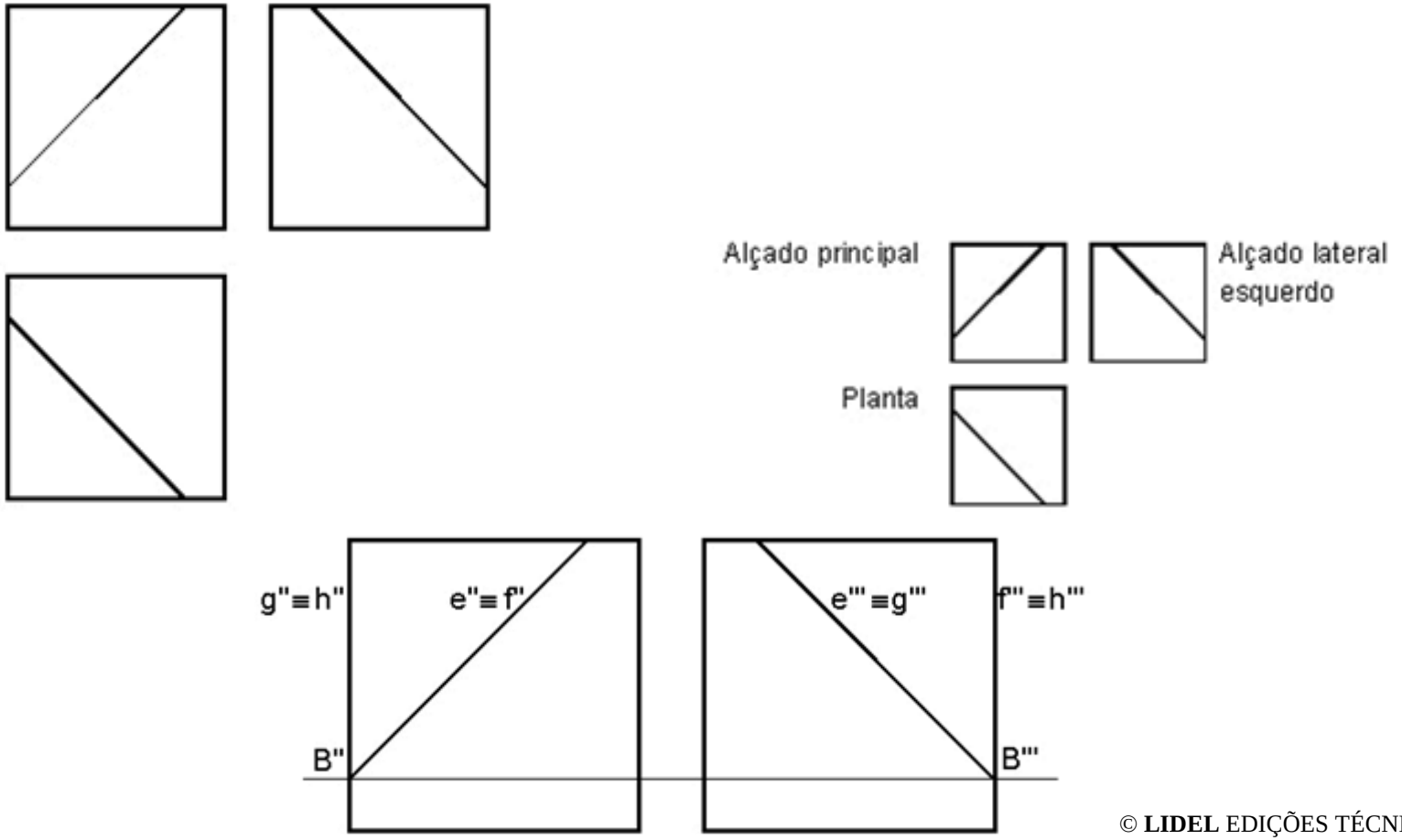
PERSPECTIVAS

Cotagem em perspectiva

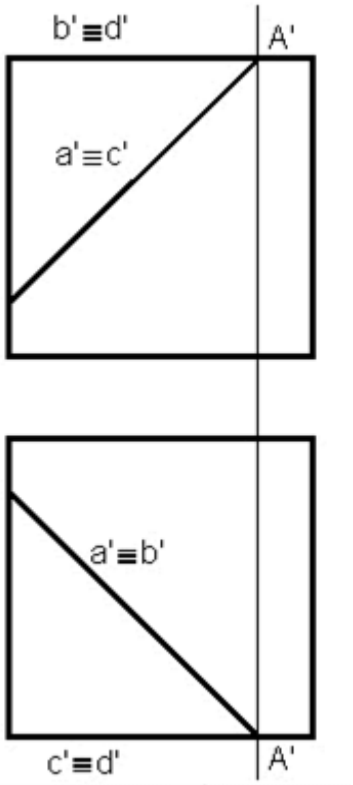


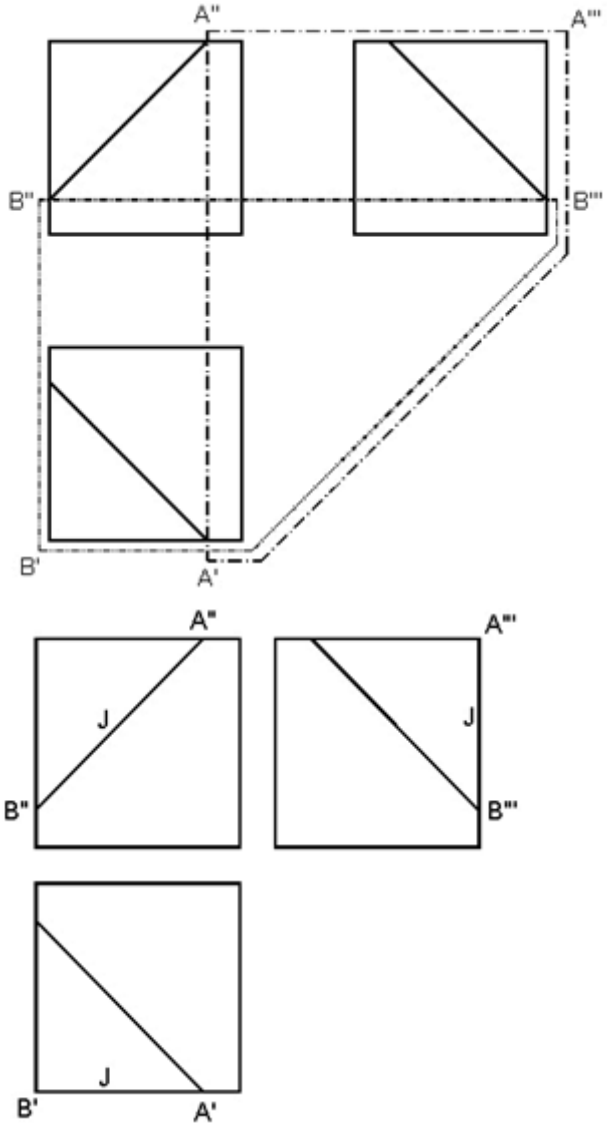
PERSPECTIVAS

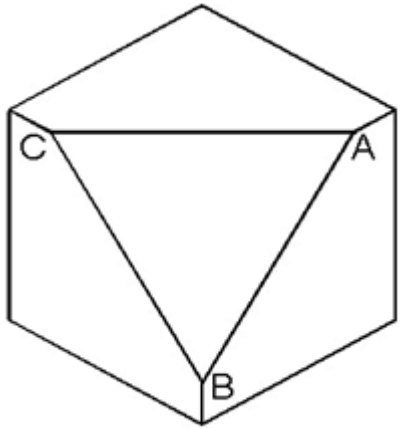
- Vistas necessárias e suficientes de uma peça, sua identificação e numeração de arestas.

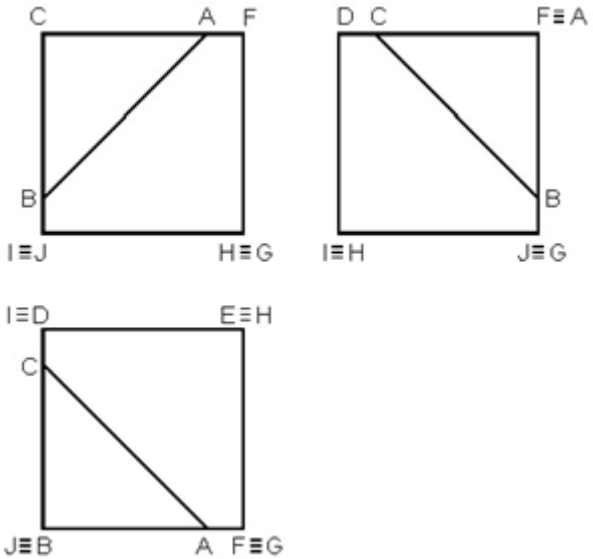


PERSPECTIVAS



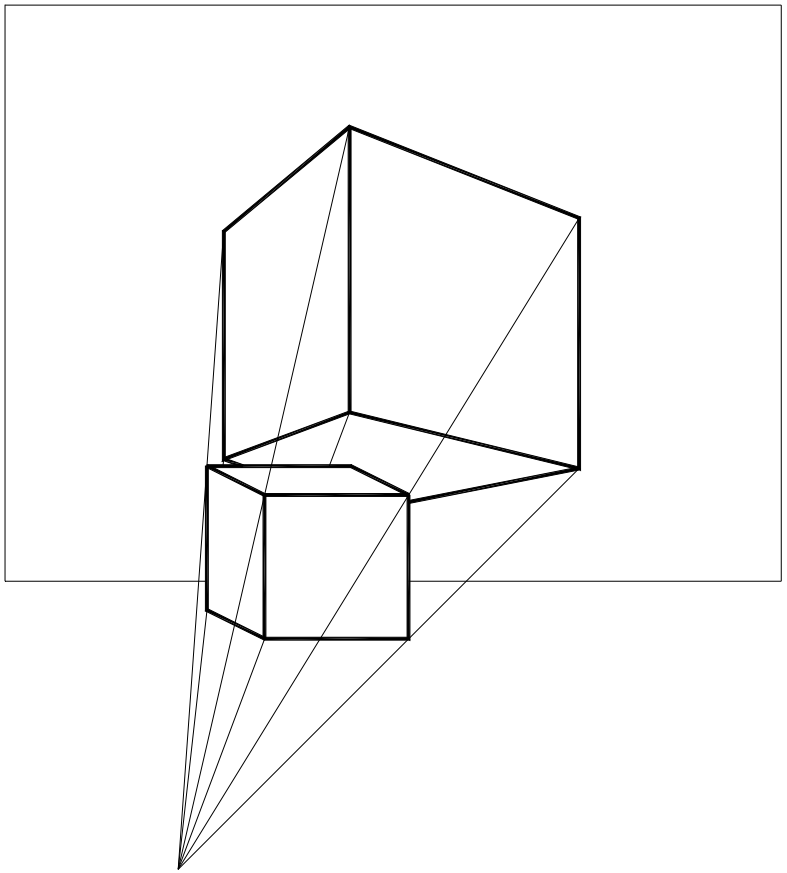


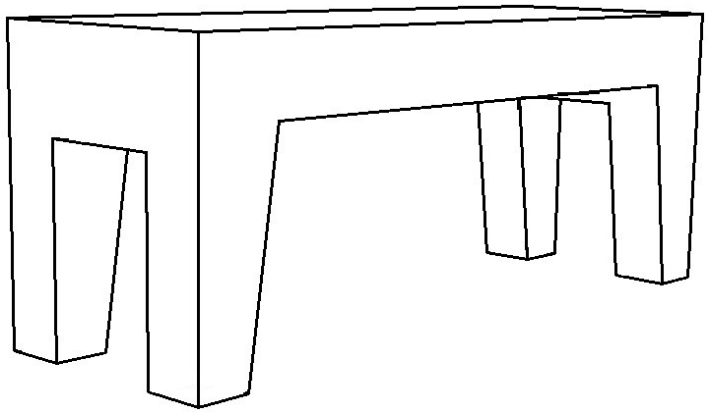




PERSPECTIVAS

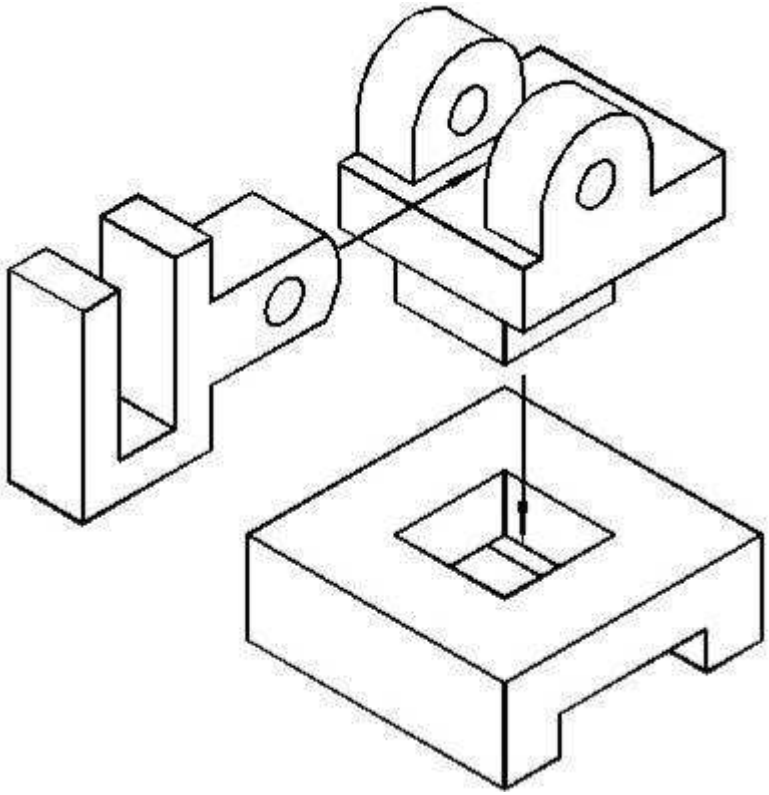
Projecções centrais

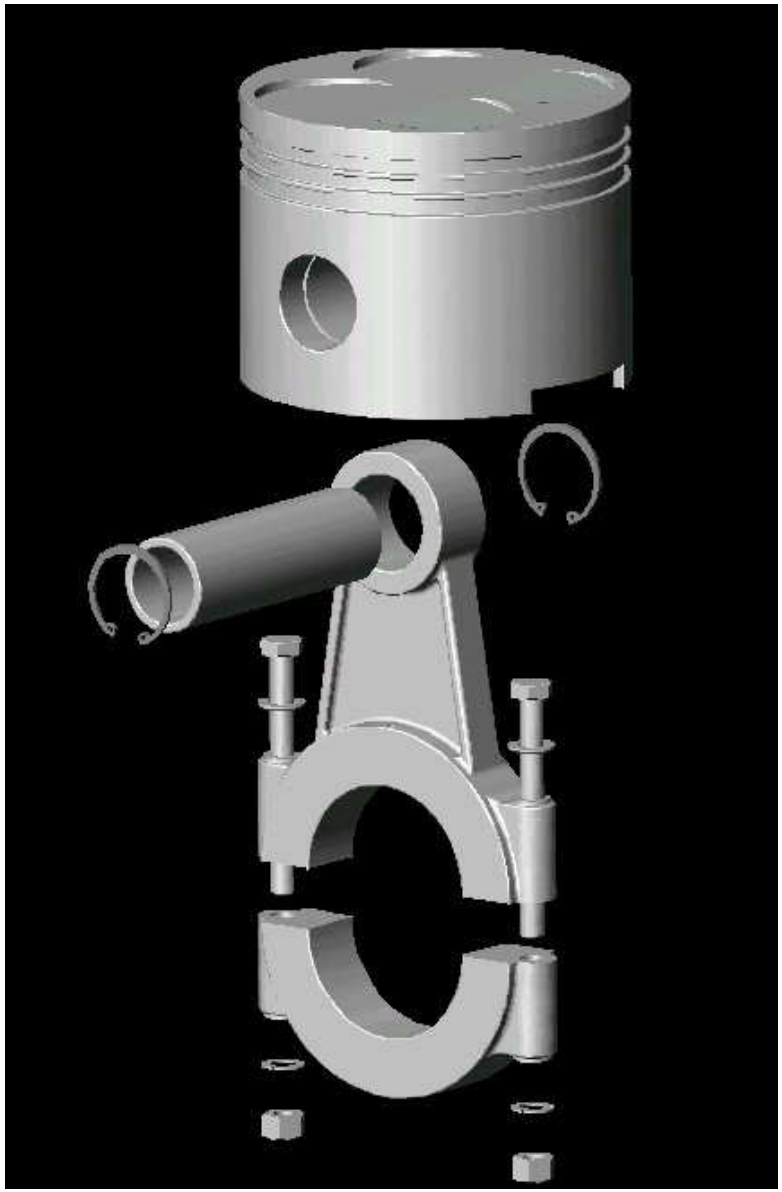




PERSPECTIVAS

Perspectiva explodida
de conjuntos de peças





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